Supplementary Material: GradTDDFT: A Differentiable Time-Dependent Density Functional Theory Framework for Training Neural Network Exchange-Correlation Functionals

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S1. SUPPLEMENTARY THEORY AND METHOD DETAILS

S1.1. Differentiable Kohn–Sham SCF

The Kohn–Sham (KS) equations for a restricted closed-shell system are

$$\hat{f}_\theta[\rho]\phi_i(\mathbf{r}) = \left[-\frac{1}{2}\nabla^2 + v_{\mathrm{ext}}(\mathbf{r}) + v_J[\rho](\mathbf{r}) + v^\theta_{\mathrm{xc}}[\rho](\mathbf{r})\right]\phi_i(\mathbf{r}) = \epsilon_i\phi_i(\mathbf{r}), \tag{S1}$$

where θ denotes the neural XC parameters, $v^\theta_{\mathrm{xc}}(\mathbf{r}) = \delta E^\theta_{\mathrm{xc}}/\delta\rho(\mathbf{r})$, and $\rho(\mathbf{r}) = 2\sum_i n_i|\phi_i(\mathbf{r})|^2$ with orbital occupations n_i normalized to one for a doubly occupied restricted orbital. In the atomic orbital (AO) basis $\{\chi_\mu\}$,

$$\mathbf{F}_\theta[\mathbf{D}]\,\mathbf{C} = \mathbf{S}\mathbf{C}\boldsymbol{\epsilon}, \tag{S2}$$

with

$$\mathbf{F}_\theta[\mathbf{D}] = \mathbf{h} + \mathbf{J}[\mathbf{D}] - \alpha_\theta\mathbf{K}[\mathbf{D}] + \mathbf{V}^\theta_{\mathrm{xc}}[\mathbf{D}] + \mathbf{V}^\theta_{\mathrm{nl}}[\mathbf{D}]. \tag{S3}$$

Here $\mathbf{h}$ is the one-electron Hamiltonian, $\mathbf{J}$ and $\mathbf{K}$ are Coulomb and exact-exchange matrices, α_θ is the scalar exchange fraction used by the nonlocal hybrid term, and $\mathbf{V}^\theta_{\mathrm{nl}}$ denotes additional nonlocal contributions generated by local-HF or PT2 neural channels when those channels are active in the SCF energy.

GradTDDFT treats molecular construction as a fixed data layer. For a given geometry and basis set, PySCF supplies

$$\mathcal{I} = \{\mathbf{h}, \mathbf{S}, (\mu\nu|\lambda\sigma), \chi_\mu(\mathbf{r}_g), \nabla\chi_\mu(\mathbf{r}_g), w_g, \mathbf{r}_g\}_{g=1}^{N_g}, \tag{S4}$$

and the differentiable JAX program updates only the density, J/K contractions, XC potential, Fock matrix, orbital coefficients, TDDFT response matrices, and losses. This distinction is important: an "integral backend" may generate $\mathcal{I}$ on CPU or GPU, but training-time gradients do not require differentiating through the external integral engine.

The SCF iteration is written as the fixed-point map

$$\mathbf{D}^{(n+1)} = \Phi_\theta(\mathbf{D}^{(n)}; \mathcal{I}), \qquad \mathbf{R}_\theta(\mathbf{D}) = \mathbf{D} - \Phi_\theta(\mathbf{D}; \mathcal{I}), \tag{S5}$$

where Φ_θ denotes Fock construction, generalized diagonalization, occupation assignment, and density reconstruction. The converged density $\mathbf{D}^*_\theta$ satisfies $\mathbf{R}_\theta(\mathbf{D}^*_\theta) = 0$. GradTDDFT implements restricted and unrestricted SCF solvers, with damping, level shifting, and DIIS-style acceleration in the reference paths.[1]

Differentiable training modes. Let $\mathcal{O}_\theta(\mathbf{D};\mathcal{I})$ denote the observable map built from the XC energy, XC potential, TDDFT response matrices, excitation energies, and oscillator strengths evaluated at density $\mathbf{D}$. For a reference target $\mathbf{y}$, define a generic loss

$$\mathcal{L}(\theta,\mathbf{D}) = \ell(\mathcal{O}_\theta(\mathbf{D};\mathcal{I}),\mathbf{y})\,. \tag{S6}$$

GradTDDFT uses three mathematically distinct training modes.

Fixed-density training. A reference density $\mathbf{D}^{\mathrm{ref}}$ is treated as an external variational point:

$$\mathcal{J}_{\mathrm{fd}}(\theta) = \mathcal{L}(\theta,\mathbf{D}^{\mathrm{ref}}), \qquad \frac{d\mathbf{D}^{\mathrm{ref}}}{d\theta} = 0, \tag{S7}$$

and therefore

$$\frac{d\mathcal{J}_{\mathrm{fd}}}{d\theta} = \left.\frac{\partial\mathcal{L}(\theta,\mathbf{D})}{\partial\theta}\right|_{\mathbf{D}=\mathbf{D}^{\mathrm{ref}}}. \tag{S8}$$

This mode updates the neural XC energy, potential, and response kernel at a fixed density while excluding the self-consistent density response.

Explicit self-consistent training. Starting from an initial density $\mathbf{D}^{(0)}$, a finite SCF trajectory is generated by

$$\mathbf{D}^{(n+1)} = \Phi_\theta(\mathbf{D}^{(n)};\mathcal{I}), \qquad n = 0,\ldots,N-1, \tag{S9}$$

and the training objective is

$$\mathcal{J}_{\mathrm{exp}}^{(N)}(\theta) = \mathcal{L}\Big(\theta,\mathbf{D}_\theta^{(N)}\Big)\,. \tag{S10}$$

The gradient follows the differentiated finite trajectory. With $\boldsymbol{\lambda}^{(N)} = \partial\mathcal{L}/\partial\mathbf{D}^{(N)}$ and

$$\boldsymbol{\lambda}^{(n)} = \left(\frac{\partial\Phi_\theta}{\partial\mathbf{D}^{(n)}}\right)^T \boldsymbol{\lambda}^{(n+1)}, \qquad n = N-1,\ldots,0, \tag{S11}$$

the parameter derivative is

$$\frac{d\mathcal{J}_{\mathrm{exp}}^{(N)}}{d\theta} = \frac{\partial\mathcal{L}}{\partial\theta} + \sum_{n=0}^{N-1}\Big(\boldsymbol{\lambda}^{(n+1)}\Big)^T \frac{\partial\Phi_\theta(\mathbf{D}^{(n)};\mathcal{I})}{\partial\theta}. \tag{S12}$$

This mode preserves the dependence of the loss on every retained SCF iterate.

Implicit self-consistent training. The SCF density is first defined as the fixed point

$$\mathbf{R}_\theta(\mathbf{D}_\theta^*) = 0, \qquad \mathbf{R}_\theta(\mathbf{D}) = \mathbf{D} - \Phi_\theta(\mathbf{D};\mathcal{I}), \tag{S13}$$

and the objective is

$$\mathcal{J}_{\mathrm{imp}}(\theta) = \mathcal{L}(\theta,\mathbf{D}_\theta^*)\,. \tag{S14}$$

The density response is represented by the adjoint variable $\boldsymbol{\lambda}$ satisfying

$$\left(\frac{\partial \mathbf{R}_\theta}{\partial \mathbf{D}}\right)^T \boldsymbol{\lambda} = -\frac{\partial \mathcal{L}}{\partial \mathbf{D}^*}, \tag{S15}$$

which gives

$$\frac{d\mathcal{J}_{\text{imp}}}{d\theta} = \frac{\partial \mathcal{L}}{\partial \theta} + \boldsymbol{\lambda}^T \frac{\partial \mathbf{R}_\theta}{\partial \theta}. \tag{S16}$$

This mode differentiates the converged SCF solution without treating the finite iteration history as part of the objective.

S1.2. Linear-Response TDDFT

Within the linear-response formalism, excitation energies Ω_I and transition amplitudes are obtained by solving the Casida eigenvalue equation:[2]

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{A}^* \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix} = \Omega_I \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\mathbf{1} \end{pmatrix} \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix}, \tag{S17}$$

where $\mathbf{X}_I$ and $\mathbf{Y}_I$ are the excitation and de-excitation amplitudes in the occupied-virtual MO rotation basis.

For restricted closed-shell singlet excitations, the orbital Hessian matrix elements are

$$\begin{aligned} A_{ia,jb} &= (\epsilon_a - \epsilon_i)\delta_{ij}\delta_{ab} + 2(ia|jb) - \alpha_\theta(ij|ab) + K^{\text{xc,A}}_{ia,jb} + K^{\text{nl,A}}_{ia,jb}, \\ B_{ia,jb} &= 2(ia|bj) - \alpha_\theta(ib|aj) + K^{\text{xc,B}}_{ia,jb} + K^{\text{nl,B}}_{ia,jb}, \end{aligned} \tag{S18}$$

where $(pq|rs)$ are two-electron integrals in the MO basis, α_θ is the effective Hartree–Fock exchange fraction, K^{xc} is the adiabatic semilocal/neural local response contribution (Section S1 S1.4), and K^{nl} collects additional nonlocal response terms from neural hybrid channels when they are enabled. Indices i, j denote occupied MOs and a, b virtual MOs. In the default semilocal response path, $K^{\text{xc,A}} = K^{\text{xc,B}}$; nonlocal channels may contribute different A and B matrices.

The Tamm–Dancoff form (TDA) sets $\mathbf{B} = 0$, reducing Eq. S17 to the Hermitian eigenvalue problem $\mathbf{A}\mathbf{X}_I = \Omega_I \mathbf{X}_I$.

From the solution vectors, physical observables are computed:

$$\boldsymbol{\mu}_I = \sum_{ia} (X_{ia,I} + Y_{ia,I}) \boldsymbol{\mu}_{ia}, \tag{S19}$$

$$f_I = \frac{2}{3} \Omega_I |\boldsymbol{\mu}_I|^2, \tag{S20}$$

where $\boldsymbol{\mu}_{ia} = \langle i|\mathbf{r}|a\rangle$ are MO transition dipole integrals, and f_I are oscillator strengths.

S1.3. Implicit Differentiation of TDDFT Roots

For self-consistent excited-state training, the response matrices depend on the neural parameters in two ways. First, the semilocal response tensor and nonlocal channel coefficients depend explicitly on θ. Second, the response problem is evaluated at the converged SCF density $\mathbf{D}^*_\theta$, whose derivative is handled by the implicit SCF adjoint in Eqs. S15–S16. The TDDFT root derivative is then taken with respect to the response operator built at this converged state. The eigensolver derivative follows implicit differentiation of a converged dominant eigensolver, so the backward pass is defined by the fixed-point/eigenpair conditions rather than by the individual Davidson iterations.[3] This is the root-energy differential used in the implementation. The Davidson iterations are used to obtain the forward Ritz amplitudes, but the operator applications inside the iterative solve and the final amplitudes are detached from the reverse pass. The differentiable root energy is then reconstructed by applying the current response operator to those fixed amplitudes.

For TDA, write the response operator as

$$\mathbf{A}_\theta \mathbf{X}_I = \Omega_I \mathbf{X}_I. \tag{S21}$$

The iterative eigensolver returns a converged Ritz vector $\widehat{\mathbf{X}}_I$. In the reverse pass, GradTDDFT treats this vector as fixed and reconstructs the differentiable excitation energy from the Rayleigh quotient

$$\Omega_I^{\mathrm{TDA}}(\theta) = \frac{\widehat{\mathbf{X}}_I^T \mathbf{A}_\theta \widehat{\mathbf{X}}_I}{\widehat{\mathbf{X}}_I^T \widehat{\mathbf{X}}_I}. \tag{S22}$$

For a scalar loss depending on TDA roots, this gives the root-energy contribution

$$\frac{d\mathcal{L}}{d\theta} \supset \sum_I \frac{\partial \mathcal{L}}{\partial \Omega_I} \frac{\widehat{\mathbf{X}}_I^T \left(d\mathbf{A}_\theta / d\theta \right) \widehat{\mathbf{X}}_I}{\widehat{\mathbf{X}}_I^T \widehat{\mathbf{X}}_I}, \tag{S23}$$

where $d\mathbf{A}_\theta/d\theta$ includes both explicit neural kernel derivatives and the derivative of the converged SCF state.

For full TDDFT, define

$$\mathbf{H}_\theta = \begin{pmatrix} \mathbf{A}_\theta & \mathbf{B}_\theta \\ \mathbf{B}_\theta & \mathbf{A}_\theta \end{pmatrix}, \qquad \mathbf{S} = \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\mathbf{1} \end{pmatrix}, \qquad \mathbf{Z}_I = \begin{pmatrix} \mathbf{X}_I \\ \mathbf{Y}_I \end{pmatrix}. \tag{S24}$$

The Casida equation is $\mathbf{H}_\theta \mathbf{Z}_I = \Omega_I \mathbf{S} \mathbf{Z}_I$. With converged Ritz amplitudes $\widehat{\mathbf{X}}_I, \widehat{\mathbf{Y}}_I$ held fixed in the reverse pass, the differentiable root is

$$\Omega_I^{\text{TDDFT}}(\theta) = \frac{\widehat{\mathbf{X}}_I^T \left(\mathbf{A}_\theta \widehat{\mathbf{X}}_I + \mathbf{B}_\theta \widehat{\mathbf{Y}}_I\right) + \widehat{\mathbf{Y}}_I^T \left(\mathbf{B}_\theta \widehat{\mathbf{X}}_I + \mathbf{A}_\theta \widehat{\mathbf{Y}}_I\right)}{\widehat{\mathbf{X}}_I^T \widehat{\mathbf{X}}_I - \widehat{\mathbf{Y}}_I^T \widehat{\mathbf{Y}}_I}. \tag{S25}$$

Therefore

$$\begin{aligned} \frac{d\Omega_I^{\text{TDDFT}}}{d\theta} = \frac{1}{N_I} \Big[& \widehat{\mathbf{X}}_I^T \left(d\mathbf{A}_\theta \widehat{\mathbf{X}}_I + d\mathbf{B}_\theta \widehat{\mathbf{Y}}_I\right) \\ & + \widehat{\mathbf{Y}}_I^T \left(d\mathbf{B}_\theta \widehat{\mathbf{X}}_I + d\mathbf{A}_\theta \widehat{\mathbf{Y}}_I\right) \Big], \qquad N_I = \widehat{\mathbf{X}}_I^T \widehat{\mathbf{X}}_I - \widehat{\mathbf{Y}}_I^T \widehat{\mathbf{Y}}_I. \end{aligned} \tag{S26}$$

Here $d\mathbf{A}_\theta$ and $d\mathbf{B}_\theta$ denote total parameter differentials through the response matrix actions, including the implicit SCF dependence. This is the mathematical form of the implementation strategy: the Davidson iterations provide the forward Ritz vectors, while the reverse-mode derivative follows the current response operator rather than the eigensolver iteration history. Amplitude-dependent observables, such as oscillator strengths, are evaluated from the same converged roots and transition dipole contractions used in the forward calculation. In the current training path, these observables use the fixed forward amplitudes returned by the eigensolver; the formulas above should therefore be interpreted as root-energy derivatives rather than full eigenvector-response derivatives for oscillator strengths.

S1.4. Energy-Consistent f_{xc} Kernel via JAX Automatic Differentiation

The exchange-correlation kernel is the second functional derivative of the XC energy with respect to the density:

$$f_{\text{xc}}(\mathbf{r}, \mathbf{r}') = \frac{\delta^2 E_{\text{xc}}}{\delta\rho(\mathbf{r})\delta\rho(\mathbf{r}')}. \tag{S27}$$

In most quantum chemistry packages, f_{xc} is implemented through analytic derivatives of standard XC functional forms or via finite differences, while automatic differentiation provides a systematic route to higher-order density-functional response derivatives.[4] For neural XC functionals, where the functional form is a neural network with thousands of parameters, analytic derivatives are infeasible and finite differences introduce numerical noise that compromises gradient-based optimization.

GradTDDFT addresses this through an *energy-consistent local response tensor*. The XC energy is expressed as an integral over a local energy contribution:

$$E_{\text{xc}}^\theta = \sum_g w_g \, e_{\text{xc}}^\theta(\mathbf{u}_g, \mathbf{q}_g), \tag{S28}$$

where $\mathbf{u}_g$ contains the differentiable density variables and $\mathbf{q}_g$ contains auxiliary frozen or separately treated features. For the current restricted MGGA response path,

$$\mathbf{u}_g = [\rho_g,\, \partial_x\rho_g,\, \partial_y\rho_g,\, \partial_z\rho_g,\, \tau_g]\,, \tag{S29}$$

with the GGA path obtained by omitting τ_g. The auxiliary features $\mathbf{q}_g$ include local exact-exchange descriptors and optional projected PT2 descriptors used by the coefficient network. For a conventional LDA, GGA, or meta-GGA functional, the same response core is obtained by taking $e^{\theta}_{\mathrm{xc}} = e^{F}_{\mathrm{xc}}$ for the selected functional F. Thus differentiable evaluation is used not only for neural functionals but also to compute the response kernel of traditional functionals from the local energy-density expression. The corresponding data flow is summarized in Fig. 1 of the main text.

The local f_{xc} contribution is constructed by contracting the grid Hessian with transition response features:

$$K^{\mathrm{xc}}_{ia,jb} = 2\sum_{g} w_g \sum_{xy} H^{\theta}_{xy,g} P^{ia}_{x,g} P^{jb}_{y,g}, \tag{S30}$$

where

$$H^{\theta}_{xy,g} = \frac{\partial^2 e^{\theta}_{\mathrm{xc}}}{\partial u_{x,g}\,\partial u_{y,g}}, \tag{S31}$$

$$P^{ia}_{x,g} = \frac{\partial u_{x,g}}{\partial \kappa_{ia}}, \tag{S32}$$

and κ_{ia} are occupied-virtual orbital rotation parameters. For example, the density feature has $P^{ia}_{\rho,g} = 2\phi_i(\mathbf{r}_g)\phi_a(\mathbf{r}_g)$ in the restricted singlet convention; gradient and kinetic-energy features are obtained by differentiating the corresponding AO-grid expressions with respect to the same orbital rotation.

Operationally, GradTDDFT treats the pointwise local energy as a scalar map

$$\mathcal{E}_\theta : \mathbf{u}_g \mapsto e^{\theta}_{\mathrm{xc}}(\mathbf{u}_g, \mathbf{q}_g), \tag{S33}$$

and evaluates $H^{\theta}_{xy,g}$ as the Hessian $\nabla^2_{\mathbf{u}}\mathcal{E}_\theta$ by automatic differentiation. The first derivative of the same scalar map gives the SCF potential components

$$v^{\theta}_{x,g} = \frac{\partial \mathcal{E}_\theta}{\partial u_{x,g}}, \tag{S34}$$

so the energy, potential, and response tensor stay internally consistent. This eliminates finite-difference errors and correctly includes derivatives of the learned mixing coefficients with respect to the density variables.

For hybrid functionals with HF exchange, the ordinary exact-exchange kernel is treated as a nonlocal contribution:

$$K_{ia,jb}^{\mathrm{HF,A}} = -\alpha_\theta (ij|ab), \qquad K_{ia,jb}^{\mathrm{HF,B}} = -\alpha_\theta (ib|aj). \tag{S35}$$

For the neural HF channel, α_θ is obtained from the density-weighted average of the learned local exchange field,

$$\alpha_x^\theta = \frac{\sum_g w_g \rho_g c_x^\theta(\mathbf{z}_g)}{\sum_g w_g \rho_g}. \tag{S36}$$

The local HF descriptor h_g remains part of the coefficient input, but it is held fixed inside the local Hessian. The orbital Hessian therefore uses Eq. S35 with $\alpha_\theta = \alpha_x^\theta$, rather than a separate response feature for the projected local HF field.

For the PT2 channel, GradTDDFT follows the CIS(D)-type double-hybrid excited-state form[5,6]. The local pair gauge p_g and the learned coefficient field define a density-weighted correlation coefficient

$$a_c^\theta = \mathrm{clip}_{[0,1]} \left[\frac{\sum_g w_g \rho_g c_2^\theta(\mathbf{z}_g)}{\sum_g w_g \rho_g} \right]. \tag{S37}$$

The linear-response problem is solved first with the semilocal kernel and the scalar HF exchange kernel. The perturbative doubles contribution is then added after the TDA or Casida eigenproblem, so the PT2 local channel does not enter the response matrix as a local second derivative. For each root,

$$\Omega_I = \Omega_I^{\mathrm{LR}} + a_c^\theta \Delta_I^{\mathrm{CIS(D)}}. \tag{S38}$$

All quantities in $\Delta_I^{\mathrm{CIS(D)}}$ are built from the current MO coefficients, orbital energies, two-electron integrals, and singles amplitudes. The correction is root-specific and is evaluated without spin-component scaling or damping in the present form.

Let i, j, k denote occupied spin orbitals, a, b, c virtual spin orbitals, and

$$\langle pq||rs\rangle = (pr|qs) - (ps|qr) \tag{S39}$$

in chemists' notation. Define the positive orbital-energy gap

$$D_{ij}^{ab} = \epsilon_a + \epsilon_b - \epsilon_i - \epsilon_j, \qquad t_{ij}^{ab} = -\frac{\langle ij||ab\rangle}{D_{ij}^{ab}}, \tag{S40}$$

where t_{ij}^{ab} is the corresponding MP2 doubles amplitude in this denominator convention. Let B_{ia}^I denote the singles transition amplitude used by the correction, with $B = X$ for TDA and $B = X + Y$ for full TDDFT. The direct CIS(D) contribution is

$$\Delta_{I,\mathrm{dir}}^{\mathrm{CIS(D)}} = -\frac{1}{4}\sum_{ijab}\frac{\left(U_{ijab}^{I}\right)^{2}}{\epsilon_a + \epsilon_b - \epsilon_i - \epsilon_j - \Omega_I^{\mathrm{LR}}}, \tag{S41}$$

where

$$\begin{aligned} U_{ijab}^{I} = &\sum_{c}\left(\langle ab||cj\rangle B_{ic}^{I} - \langle ab||ci\rangle B_{jc}^{I}\right) \\ &+ \sum_{k}\left(\langle ka||ij\rangle B_{kb}^{I} - \langle kb||ij\rangle B_{ka}^{I}\right). \end{aligned} \tag{S42}$$

The denominator in Eq. S41 is $D_{ij}^{ab} - \Omega_I^{\mathrm{LR}}$; the excitation energy shifts the doubles manifold relative to the singles state.

The indirect term accounts for the coupling of the CIS singles vector with ground-state MP2 amplitudes. The intermediates used in GradTDDFT are

$$R_{ab} = -\sum_{jkc}(jc|kb)\, t_{jk}^{ca}, \tag{S43}$$

$$R_{ij} = -\sum_{kab}(ja|kb)\, t_{ik}^{ab}, \tag{S44}$$

$$W_{ia}^{I} = \sum_{jkbc}\langle jk||bc\rangle\, t_{ik}^{ac}\, B_{jb}^{I}. \tag{S45}$$

The unscaled indirect correction is then

$$\begin{aligned} \Delta_{I,\mathrm{ind}}^{\mathrm{CIS(D)}} = &\sum_{iab} B_{ia}^{I} B_{ib}^{I} R_{ab} + \sum_{ijc} B_{ic}^{I} B_{jc}^{I} R_{ij} \\ &+ \sum_{ia} B_{ia}^{I} W_{ia}^{I}. \end{aligned} \tag{S46}$$

The total second-order root correction is

$$\Delta_I^{\mathrm{CIS(D)}} = \Delta_{I,\mathrm{dir}}^{\mathrm{CIS(D)}} + \Delta_{I,\mathrm{ind}}^{\mathrm{CIS(D)}}, \tag{S47}$$

and the learned double-hybrid scaling is applied only through a_c^{θ} in Eq. S38. This makes the semilocal/HF response kernel and the perturbative doubles correction separate mathematical objects while keeping both differentiable with respect to the neural functional parameters.

S1.5. Neural XC Functional Architecture

The neural XC functional in GradTDDFT follows the *coefficient-basis* paradigm used by GradDFT-style neural functionals. A pointwise neural network maps local descriptors to mixing coefficients, and those coefficients multiply a fixed set of energy-density channels. At grid point g,

$$e_{\mathrm{xc}}^{\theta}(g) = \sum_{k=1}^{n_{\mathrm{sl}}} c_k^{\theta}(\mathbf{z}_g)\, e_k^{\mathrm{sl}}(g) + \mathbb{1}_{\mathrm{pt2}}\, c_{\mathrm{pt2}}^{\theta}(\mathbf{z}_g)\, p_g + c_{\mathrm{hf}}^{\theta}(\mathbf{z}_g)\, h_g, \tag{S48}$$

and

$$E_{\mathrm{xc}}^{\theta} = \sum_{g} w_g e_{\mathrm{xc}}^{\theta}(g). \tag{S49}$$

The channel densities in Eq. S48 are defined on the same quadrature grid. A semilocal channel has the general spin-resolved meta-GGA form

$$E_k^{\mathrm{sl}} = \int e_k^{\mathrm{sl}}(\mathbf{r})\, d\mathbf{r}, \qquad e_k^{\mathrm{sl}}(\mathbf{r}) = \rho(\mathbf{r})\, \varepsilon_k^{\mathrm{sl}}(\rho_\alpha, \rho_\beta, \sigma_{\alpha\alpha}, \sigma_{\alpha\beta}, \sigma_{\beta\beta}, \tau_\alpha, \tau_\beta)_{\mathbf{r}}\,, \tag{S50}$$

where $\rho = \rho_\alpha + \rho_\beta$, $\sigma_{\sigma\sigma'} = \nabla\rho_\sigma \cdot \nabla\rho_{\sigma'}$, and τ_σ is the spin kinetic-energy density. GGA channels are obtained by omitting τ_σ, and LDA channels by omitting both gradient and kinetic-energy variables. On the grid, $E_k^{\mathrm{sl}} \simeq \sum_g w_g e_k^{\mathrm{sl}}(g)$.

The semilocal channel set is not restricted to the B3LYP-like choice below. Through jax-xc,[7] translated Libxc LDA, GGA, and meta-GGA energy densities can be inserted as channels,

$$\mathcal{B}_{\mathrm{sl}} = \left\{ e_{\mathrm{xc}}^{F_1}(g), \ldots, e_{\mathrm{xc}}^{F_M}(g) \right\}, \tag{S51}$$

where each F_m denotes a conventional semilocal functional form. A fixed functional mixture is recovered by constant coefficients a_m,

$$e_{\mathrm{xc}}^{F}(g) = \sum_{m=1}^{M} a_m e_{\mathrm{xc}}^{F_m}(g), \tag{S52}$$

whereas GradTDDFT replaces a_m by learned pointwise fields $c_m^{\theta}(\mathbf{z}_g)$. This gives neural-network versions of different semilocal functional families without changing the SCF or TDDFT response equations.

The HF channel is a projected local representation of the exact-exchange energy, following the same local-energy-density viewpoint used in doubly local double-hybrid functionals.[8] In spin-orbital notation,

$$E_x^{\mathrm{HF}} = -\frac{1}{2} \sum_{\sigma} \sum_{ij}^{\mathrm{occ}} n_{i\sigma} n_{j\sigma} \iint \frac{\phi_{i\sigma}(\mathbf{r})\phi_{j\sigma}(\mathbf{r})\phi_{j\sigma}(\mathbf{r}')\phi_{i\sigma}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d\mathbf{r}\, d\mathbf{r}'. \tag{S53}$$

The local descriptor used in Eq. S48 is the exchange-hole-gauge density

$$h_g = -\frac{1}{2} \sum_{\sigma} \sum_{ij}^{\mathrm{occ}} n_{i\sigma} n_{j\sigma} \phi_{i\sigma}(\mathbf{r}_g)\phi_{j\sigma}(\mathbf{r}_g) v_{ij\sigma}^{x}(\mathbf{r}_g), \qquad v_{ij\sigma}^{x}(\mathbf{r}) = \int \frac{\phi_{j\sigma}(\mathbf{r}')\phi_{i\sigma}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d\mathbf{r}', \tag{S54}$$

so that $\sum_g w_g h_g$ is the quadrature representation of E_x^{HF} up to grid and projection error.

The optional PT2 channel uses the same local-channel construction. It starts from the canonical second-order correlation expression

$$E_c^{(2)} = \frac{1}{4} \sum_{ij}^{\mathrm{occ}} \sum_{ab}^{\mathrm{virt}} \frac{|\langle ij||ab\rangle|^2}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}, \qquad \langle ij||ab\rangle = (ij|ab) - (ij|ba), \tag{S55}$$

where i, j are occupied spin orbitals and a, b are virtual spin orbitals. GradTDDFT uses a local projected density p_g satisfying

$$E_c^{(2)} \simeq \sum_g w_g p_g, \qquad p_g = \frac{1}{4} \sum_{ijab} t_{ij}^{ab}\, \eta_{ij}^{ab}(\mathbf{r}_g), \qquad t_{ij}^{ab} = \frac{\langle ij||ab\rangle}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}, \tag{S56}$$

with η_{ij}^{ab} denoting the chosen real-space projection of the antisymmetrized pair interaction, normalized so that $\sum_g w_g \eta_{ij}^{ab}(\mathbf{r}_g) \simeq \langle ij||ab\rangle$. Thus the neural coefficient multiplies a pointwise PT2 energy density while the underlying orbital-energy denominators and pair interactions retain their standard perturbative meaning.

The semilocal channels e_k^{sl} are evaluated by the JAX/libxc-compatible backend. The default B3LYP-like basis is

$$\{e_x^{\mathrm{LDA}}, e_x^{\mathrm{B88}}, e_c^{\mathrm{VWN\text{-}RPA}}, e_c^{\mathrm{LYP}}\}, \tag{S57}$$

with coefficient priors matching the B3LYP local decomposition and a separate HF prior. If the PT2 channel is enabled, it is inserted between the semilocal channels and the HF channel with a zero initial prior.

Input descriptors. The model supports two pointwise input families. The canonical input uses a spin-resolved descriptor:

$$\mathbf{z}_g^{\mathrm{can}} = [\rho_\alpha, \rho_\beta, \sigma, \sigma_{\alpha\alpha}, \sigma_{\beta\beta}, \tau_\alpha, \tau_\beta, h_\alpha^{\omega_1}, \ldots, h_\alpha^{\omega_m}, h_\beta^{\omega_1}, \ldots, h_\beta^{\omega_m}, p]_g, \tag{S58}$$

where $\sigma_{\alpha\beta} = \nabla\rho_\alpha \cdot \nabla\rho_\beta$, τ is the kinetic energy density, h^ω are local range-separated HF descriptors, and p is the optional PT2 local descriptor. The default implementation uses $m = 2$ HF descriptor channels when local HF features are available. The enhanced input augments these variables with ρ, $\log(1+\rho)$, $\sqrt{\rho}$, the full set of spin-gradient invariants, total τ, and a semilocal energy-density descriptor. These descriptors are all computed on the PySCF quadrature grid but enter the training graph as JAX arrays.

Pointwise mixing network. Let $\mathbf{a}_g^\theta = N_\theta(\mathbf{z}_g)$ be the raw network output. The final channel coefficients are bounded as

$$c_g^\theta = s \text{ sigmoid}\left(\frac{\mathbf{a}_g^\theta}{s}\right), \tag{S59}$$

with scale $s = 2$ by default. Semilocal coefficients are then clipped to the configured kernel range, while HF and PT2 heads are clipped to the unit interval before they are used as mixing fields. The output dimension is

$$n_{\text{out}} = n_{\text{sl}} + 1 + \mathbb{1}_{\text{pt2}}, \tag{S60}$$

corresponding to semilocal channels, optional PT2, and HF.

GradTDDFT provides two MLP architectures. The default residual network applies

$$\mathbf{x}_0 = \tanh(W_0 \log(|\mathbf{z}| + \epsilon) + b_0)\,, \qquad \mathbf{x}_{\ell+1} = \varphi(\text{LayerNorm}(W_\ell \mathbf{x}_\ell + \mathbf{b}_\ell + \Pi_\ell \mathbf{x}_\ell))\,, \tag{S61}$$

where Π_ℓ is either the identity or a learned projection when layer widths change. The default hidden width is six layers of 256 units, and the default residual-block activation is ELU. The lighter simple MLP instead applies the smooth transform $\frac{1}{2}\log(\mathbf{z}^2 + \epsilon^2)$ followed by dense layers and a configurable activation. Both networks are implemented in JAX/Flax.[9]

HF and PT2 heads. The HF channel h_g is a projected local exact-exchange contribution and the PT2 channel p_g is a projected second-order perturbative local contribution. The HF field produces both a local coefficient and an effective scalar exchange fraction

$$\alpha_{\text{eff}}^\theta = \frac{\sum_g w_g\, \rho_g\, c_{\text{hf}}^\theta(\mathbf{z}_g)}{\sum_g w_g\, \rho_g}, \tag{S62}$$

which enters Eq. S35. In the excited-state response, the local HF descriptor field is treated as a fixed coefficient input for the local Hessian, and the nonlocal TDDFT response uses the scalar exchange fraction in Eq. S62. The PT2 head defines the scalar coefficient in Eq. S37 for the CIS(D)-type excited-state correction.

S1.6. Differentiable Objective Terms

The differentiable formulation supports several classes of scalar objectives. Rather than prescribing a fixed training sequence, we write the objective as a weighted combination of ground-state, and excited-state terms:

$$\mathcal{J}(\theta) = \lambda_{\text{GS}} \mathcal{L}_{\text{GS}}(\theta) + \lambda_{\text{ES}} \mathcal{L}_{\text{ES}}(\theta), \tag{S63}$$

where the weights determine which physical constraints are active in a given experiment.

Ground-state terms. Ground-state reference data can enter through total energies, densities, or both:

$$\mathcal{L}_{\mathrm{GS}} = w_E \cdot \mathrm{MSE}(E^\theta, E^{\mathrm{ref}}) + w_\rho \cdot \mathrm{MAE}(\rho^\theta, \rho^{\mathrm{ref}}), \tag{S64}$$

where E^θ is the SCF total energy obtained with the neural functional and ρ^θ is the corresponding self-consistent density. Depending on the chosen differentiable mode in Eqs. S7–S16, the density may be fixed, propagated through a finite SCF trajectory, or treated as the solution of an implicit fixed-point problem.

Excited-state terms. Excited-state reference data enter through vertical excitation energies and oscillator strengths:

$$\mathcal{L}_{\mathrm{ES}} = w_\Omega \cdot \mathrm{MAE}(\Omega_I^\theta, \Omega_I^{\mathrm{ref}}) + w_f \cdot \mathrm{MAE}(f_I^\theta, f_I^{\mathrm{ref}}), \tag{S65}$$

where Ω_I and f_I are the excitation energies and oscillator strengths for selected excited states I. The derivative of this loss includes the dependence of the local energy density e_{xc}^θ, XC potential components, local f_{xc} tensor, orbital Hessian matrices, Casida eigenvectors, and oscillator strengths on the neural parameters.

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